PAPER_280: Saturn UQFF Solar Tidal Perturbation Ratio τ_Sun

Session: 78
Module: SATURN_UQFF_MODULE.cpp (21st C++ module — first planetary-scale UQFF module)
New Constants: τ_Sun (Solar UQFF Tidal Perturbation Ratio), g_Sun_tidal (Solar tidal surface acceleration)
Status: UNIQUE — establishes the UQFF framework for Solar System planetary bodies

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Abstract

Saturn is the first planetary-scale body in the UQFF module catalogue. All prior 20 C++ modules described stellar objects, neutron stars, or galaxies. The transition to a Solar System planet (z=0, f_DM=0, r=6.0268×10^{7} m) introduces a class of external gravitational coupling absent from stellar/galactic UQFF modules: the tidal perturbation exerted by the host star (the Sun) on the planet's surface gravity field. This paper defines and derives the **Solar UQFF Tidal Perturbation Ratio** τ_Sun, a dimensionless constant that quantifies the ratio of the Sun's tidal acceleration at Saturn's orbit to Saturn's own surface gravity. The ratio is 6.22×10^{-6} — small but non-zero, and physically the first UQFF solar coupling constant. A universal formula is derived for any planet.

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2. Physical Motivation

For stellar and galactic UQFF modules, all gravitational contributors were internal to the system or cosmological. For a planetary body orbiting a star, a new type of perturbation exists: the **tidal gravitational acceleration** of the host star felt at the planet's surface. This is not the same as the star's direct gravity at the planet's centre (which is balanced by orbital inertia), but rather the **differential** gravitational acceleration across the finite body of the planet — the tidal component that modifies the surface gravity field.

In the UQFF framework, we model the host-star tidal term as an additive constant to g_total:

$$g_{\text{Sun\_tidal}} = \frac{G \cdot M_{\odot}}{r_{\text{orbit}}^2}$$

This is the raw solar gravitational acceleration at Saturn's orbital radius — the scale of the tidal perturbation at the planet's orbital position.

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3. Derivation

3.1 Saturn Surface Gravity (g_base)

$$g_{\text{base}} = \frac{GM_{\text{Saturn}}}{r_{\text{Saturn}}^2} = \frac{6.674\times10^{-11}\times5.683\times10^{26}}{(6.0268\times10^{7})^2}$$

$$g_{\text{base}} = \frac{3.793\times10^{16}}{3.632\times10^{15}} = \mathbf{10.44}\ \text{m/s}^{\mathbf{2}}$$

Note: g_base = 10.44 m/s^2 is 14 orders of magnitude larger than typical galactic g_base (~10^{-10} m/s^2). This is the first UQFF module where pre_sum_Ug = 52 × g_base = 542.9 m/s^2 > 1 m/s^2.

3.2 Solar Tidal Acceleration at Saturn's Orbit

$$g_\odot = \frac{GM_\odot}{r_{\text{orbit}}^2} = \frac{6.674 \times 10^{-11} \times 1.989 \times 10^{30}}{(1.43 \times 10^{12})^2}$$

$$g_\odot = \frac{1.327 \times 10^{20}}{2.045 \times 10^{24}} = \mathbf{6.49 \times 10^{-5}}\ \mathrm{m/s^2}$$

3.3 Solar UQFF Tidal Perturbation Ratio

The dimensionless ratio of Sun's tidal acceleration to Saturn's surface gravity:

$$\tau_\odot = \frac{g_\odot}{g_{\text{base}}} = \frac{GM_\odot/r_{\text{orbit}}^2}{GM_{\text{Saturn}}/r_{\text{Saturn}}^2} = \frac{M_\odot}{M_{\text{Saturn}}} \cdot \left(\frac{r_{\text{Saturn}}}{r_{\text{orbit}}}\right)^2$$

$$\tau_\odot = \frac{1.989 \times 10^{30}}{5.683 \times 10^{26}} \times \left(\frac{6.0268 \times 10^7}{1.43 \times 10^{12}}\right)^2 = 3502 \times 1.776 \times 10^{-9}$$

$$\boxed{\tau_\odot = 6.22 \times 10^{-6}}$$

3.4 Universal Planetary Formula

For any planet orbiting any star:

$$\tau_{\text{planet}} = \frac{M_{\text{star}}}{M_{\text{planet}}} \cdot \left(\frac{r_{\text{planet}}}{r_{\text{orbit}}}\right)^2$$

This UQFF ratio governs the strength of the host-star tidal perturbation on the planet's surface gravity field in the UQFF framework.

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4. Results — Solar System Comparison

Planet	M_planet (kg)	r_planet (m)	r_orbit (m)	τ_Sun
Mercury	3.30×10^{23}	2.44×10^{6}	5.79×10^{10}	$\mathbf{1.07\times10^{-2}}$
Earth	5.97×10^{24}	6.37×10^{6}	1.496×10^{11}	$\mathbf{6.03\times10^{-4}}$
Jupiter	1.898×10^{27}	7.15×10^{7}	7.78×10^{11}	$\mathbf{8.84\times10^{-6}}$
Saturn	$\mathbf{5.683\times10^{26}}$	$\mathbf{6.0268\times10^{7}}$	$\mathbf{1.43\times10^{12}}$	$\mathbf{6.22\times10^{-6}}$

Trend: τ decreases with increasing orbital radius and planet mass. Mercury's τ = 0.0107 (solar tidal is ~1% of surface gravity). Saturn's τ = 6.22×10^{-6} (parts-per-million perturbation).

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5. Integration in computeG()

In SATURN_UQFF_MODULE.cpp, g_Sun_tidal enters as a constant additive term:

```
g_total = [g_grav + Ug_sum + Lambda + quantum + Lorentz + fluid
           + F_ring_tidal(t) + g_Sun_tidal + g_exp + a_wind] × corr_SC
```

The `g_Sun_tidal` term is **constant** (not oscillatory) because at Saturn's orbital radius the Sun's tidal field is quasi-static over observational timescales. This contrasts with the ring tidal term (PAPER_281) which oscillates at ω_ring_kep.

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6. WOLFRAM_TERM Registration

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WOLFRAM_TERM_SATURN_SOLAR: "SaturnUQFF:tau_Sun=M_Sun/M*(r/r_orbit)^2=6.22e-6;
                       g_Sun_tidal=G*M_Sun/r_orbit^2=6.49e-5 m/s^2 [PAPER_280]"
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7. Significance

- **First UQFF Solar tidal coupling** — establishes UQFF Solar System framework
- **First planetary-scale module** — g_base = 10.44 m/s^2 (vs $\sim 10^{-10}$ for galaxies)
- **Universal formula** applicable to any planet around any star
- τ**_Sun = 6.22×10^{-6}** is the characteristic scale of Sun-Saturn tidal interaction in UQFF